WEEK 2 / LESSON 4 / 50 MINUTES

L4

Data Cleaning

A Myntra analyst once dropped 12 lakh "duplicate" rows that turned out to be legitimate re-orders. Revenue was under-reported for a week. Cleaning is decision-making — not pattern-matching.

What will you log the next time you call df.drop_duplicates?

WHAT'S INSIDE

WHY THIS LESSON MATTERS

12 lakh "duplicates" — all real customers.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1. Detect and count missing values per column.
2. Decide between dropna, fillna, interpolate with rationale.
3. Convert columns to the correct dtype.
4. Detect and remove only the RIGHT duplicates.
5. Produce a cleaning log a reviewer can audit.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01
NaN is not zero
The first concept to internalise

02
Three strategies — pick...
Never silently dropna() a whole frame

03
dtype conversion — the...
Converting 1M-row category from object usually cuts memory 90%+ and speeds groupby 3–5×.

04
Duplicates — decide before...
Ask: are these duplicates real or bad data?

01

PART 1 OF 4

NaN is not zero

The first concept to internalise

CONCEPT 1 OF 4

NaN is not zero

NaN means "unknown". It is NOT 0.
NaN + 1 = NaN. NaN == NaN is False.
In pandas, NaN is a float by default.
Need a missing int? Use the capital-I "Int64" dtype.
WARNING
NaN ≠ 0

Silent NaNs ruin averages, counts, and joins.

DEFINITION

NaN is not zero

The first concept to internalise

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

Three strategies — pick with intent

Drop rows: when a column is critical AND missing rate is low.
Impute: continuous column with missing-at-random — use median or mean.
Interpolate: time-ordered data — prices, sensor readings.
Leave + flag: if missingness is itself a signal (e.g. spouse_name).
SENIOR HABIT
Log every drop

Silent cleaners are not trusted with production data.

KEY POINTS

- Drop rows: when a column is critical AND missing rate is low
- Impute: continuous column with missing-at-random — use median or mean
- Interpolate: time-ordered data — prices, sensor readings
- Leave + flag: if missingness is itself a signal (e.g. spouse_name)

CONCEPT 3 OF 4

dtype conversion — the quiet productivity win

Converting 1M-row category from object usually cuts memory 90%+ and speeds groupby 3–5×.

DEFINITION

dtype conversion — the quiet...

Converting 1M-row category from object usually cuts memory 90%+ and speeds groupby 3–5×.

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

Duplicates — decide before you drop

df.duplicated().sum() — fully identical rows.
df.duplicated(subset=["order_id"]).sum() — duplicates on a business key.
Two real orders of the same item are not dupes — do not drop them.
df.drop_duplicates(subset=["order_id"], keep="first") — targeted drop.
RULE
Why before drop

"They look duplicated" is never enough.

KEY POINTS

- df.duplicated().sum() — fully identical rows
- df.duplicated(subset=["order_id"]).sum() — duplicates on a business key
- Two real orders of the same item are not dupes — do not drop them
- df.drop_duplicates(subset=["order_id"], keep="first") — targeted drop

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK · PAIR · SHARE

ACTIVITY

Drop or impute — defend your choice

PROMPT

Given: 0.8% missing in "email"; 42% missing in "spouse_name"; 5% missing in "salary". Propose a strategy for each.

HOW TO RUN IT

1. Given: 0.8% missing in "email"; 42% missing in "spouse_name"; 5% missing in "salary"....
2. Defend each choice in one sentence.
3. Room votes; facilitator moderates.
4. Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1	*A cleaning decision you regret making silently in the past?*
Q2	*Which of the four cleaning strategies will you default to this week?*
Q3	*Will you log your drops from now on? Why or why not?*

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

KNOWLEDGE CHECK · QUESTION 1 OF 3

NaN == NaN evaluates to:

- A True
- B False
- C NaN
- D Error

ANSWER · QUESTION 1 OF 3

NaN == NaN evaluates to:

CORRECT ANSWER

B

False

WHY

NaN is never equal to anything, including itself. Use .isna() to detect

KNOWLEDGE CHECK · QUESTION 2 OF 3

pd.to_numeric(s, errors="coerce") on ["1","2","abc"] returns:

A [1,2,"abc"]

B [1,2,NaN]

C Error

D [1,2,0]

ANSWER · QUESTION 2 OF 3

pd.to_numeric(s, errors="coerce") on ["1","2","abc"] returns:

CORRECT ANSWER

B

[1,2,NaN]

WHY

coerce → bad values become NaN; numeric ones parse normally

KNOWLEDGE CHECK · QUESTION 3 OF 3

Converting 1M-row object city column → category typically:

A Doubles memory

B Saves ~10%

C Saves ~90%+

D No change

ANSWER · QUESTION 3 OF 3

Converting 1M-row object city column → category typically:

CORRECT ANSWER

C

Saves ~90%+

WHY

Category stores an integer code per row + a single lookup table of unique values

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

NaN means "unknown". It is NOT 0.
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In pandas, NaN is a float by default.
Need a missing int? Use the capital-I "Int64" dtype.
WARNING
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Silent NaNs ruin averages, counts, and joins.

THE TAKEAWAY

"Silent cleaners are not trusted with production data. The log is your proof of intent."

WHAT TO NOTICE

1. NaN is not zero
2. Three strategies — pick...
3. dtype conversion — the...
4. Duplicates — decide before...

APPLY IT → ASSIGNMENT

Mini assignment — 5-step clean

Take any messy CSV (Kaggle Titanic if you need one). Clean it in five intentional steps.

DELIVERABLES

- Count missing per column.
- Decide: drop / impute / interpolate / flag for each.
- Coerce dtypes.
- Dedup on a business key.
- Keep a cleaning log at the top of the notebook.

WHAT 'GOOD' LOOKS LIKE

Deliverable: Notebook URL + cleaning log in the README.

ONE THING TO REMEMBER

"Silent cleaners are not trusted with production data. The log is your proof of intent."

— AI Learning Hub

END OF LESSON 2

What you can now do

Each one of these is now part of your working vocabulary.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.

NEXT → Lesson 5 — Joins, GroupBy, Pivot. Where clean data becomes business answers.